EXECUTIVE SUMMARY

Finding the cuts, then keeping the movie

Black-frame detection, review, and high-quality compile

23 October 2026

What this is

Broadcast recordings often put a moment of black between the movie and a commercial, or between two programs joined into one file. These scripts find those black frames, let a person throw away the pieces they do not want, and build one high-quality file from what is left.

The review copies are small and disposable. The final file is encoded from the original, using the timestamps of the previews that were not deleted. Nothing decides on its own which segment is a commercial. That choice stays with the person watching the previews.

How it works now

1. **Detect**
   h264BlackFrameExtract reads the video once. FFmpeg reports each stretch of black. The cut is the middle of that stretch. The recording is copied, not moved, into a work folder named after the file.

2. **Preview**
   Every piece between cuts is written as a 180p preview. The start and end time are in the filename. A long movie often yields a few long pieces and a run of short ones at the head and tail, where bumpers and breaks usually sit.

3. **Choose**
   Delete any preview that should not be in the finished movie. A missing preview is a rejected cut. A preview still in the folder is kept, even if playback of that preview was interrupted. The filename is the cut list, not the length of the preview file.

4. **Compile**
   h264BlackFrameCompile or h265BlackFrameCompile encodes only the surviving ranges from the copied original, in order, then joins them into one file beside that original. Audio is copied, not re-encoded.

The scripts

h264BlackFrameExtract

One recording in, one work folder out: the source copy, a black-frame log, a clip list, the 180p previews, and a launcher.

h264BlackFrameExtract-batch

Runs the extractor on every .mp4 in the current directory, one after another. It does not look in subfolders.

h264BlackFrameCompile

Run in the work folder, or point it at that folder. Writes <name>.h264.mp4 from the previews that are still there.

h265BlackFrameCompile

The same compile, with the HEVC settings below. Writes <name>.h265.mp4 and does not replace the H.264 file.

The launcher in the work folder

The .hq.sh file does not encode. It starts the H.264 compile. Older copies that demanded two arguments are out of date.

What changed

The 2018–2020 script treated FFmpeg as single-threaded. It opened a terminal per clip, pinned each encode to one thread, and slept when the CPU looked busy. It also scraped a frame-by-frame black log. The current scripts let FFmpeg use every core, run one encode at a time on the 8-core Xeon E5-2680, and take black frames from a single detection pass. The result that matters: the high-quality file is built only from the previews that remain.

Picture settings

This CPU has no hardware video encoder. Both compiles are software encodes, one clip at a time, using all 16 threads.

	H.264	H.265
Command	h264BlackFrameCompile	h265BlackFrameCompile
Encoder	libx264, slow, film tune	libx265, medium, 10-bit
Quality	CRF 18	CRF 20, about the same picture
Why	Best useful step on this CPU	A slower HEVC preset is only a few frames per second
Output	<name>.h264.mp4	<name>.h265.mp4

What a person actually runs

Point the extractor at a recording. Open the work folder, watch the previews, and delete the ones that are not part of the movie. Then run either compile script in that folder. The result is one file: the kept ranges, in order, taken from the original picture.

The previews are a cut list with pictures attached. Deleting a preview is the edit. The compile step does not guess.